Updated October 2026

Clay Swackhamer, PhD, PE

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Penn State University *2026-present*
Position: Assistant Professor
Department: Food Science
Research interests:

- Determining how food structure and texture influence oral processing and digestion
- Developing an oral processing simulator for integration with in vitro digestion workflows
- Determining how food processing alters oral food breakdown, eating behavior, and sensory experience
- Advancing our understanding of the role of the mouth or gut microbiome in health and disease

Education and training

Oregon State University *2025-2026*
Position: Research Associate
Departments: Food Science, Nutrition Science
Supervisor: Dr. David Dallas
Project: Adapting SHIME as a model for upper gastrointestinal digestion of milk under conditions representative of infant digestion.

Purdue University *2022-2025*
Position: Postdoctoral Researcher
Department: Food Science
Supervisor: Dr. Bruce Hamaker
Project: USDA-NIFA postdoctoral fellowship (2 years) to explore how debranching and cross-linking processes modify the effect of arabinoxylans extracted from maize bran on the taxonomic composition and short-chain fatty acid production of the gut microbiome.

University of California, Davis *Ph.D. 2022*
Major: Biological Systems Engineering
Minor: Modeling and Control of Biological Systems
Specialization: Food Engineering
Supervisor: Dr. Gail Bornhorst
Dissertation: Breakdown of solid foods during in vitro gastric digestion
Summary: Developed model solid foods based on the macronutrient composition of the Standard American Diet, with systematically varied softening rates, to study mechanisms of food breakdown during in vitro gastric digestion with simulated peristalsis.
GPA: 3.95

Penn State University *B.S. 2015*
Major: Biological Engineering
Minor: Spanish
Specialization: Food and Biological Process Engineering
Schreyer Honors College Scholar
GPA: 3.92

University of Alicante *Fall 2014*
Semester in Alicante, Spain
Completed 15 credits of courses taught in Spanish
GPA: 4.00

Professional Licensure

- **Professional Engineer (PE)**. Agricultural engineer registered with California Board for Professional Engineers, Land Surveyors, and Geologists. License number 589. *2022-present.*

Peer-Reviewed Publications

1. M. Biondi Ryan, **C. Swackhamer**, S. Donovan, J. Smilowitz, J. Kim, C. Stewart, D. Kuehn, G. Rauner, K. Hettinga, E. Kearns, K. Smolen, S. Reyes, G. Rai, M. McGuire, C. Russo, D. Dallas✉. "Human Milk as a Model for Next-Generation Infant Formula: Opportunities and Challenges". 2026. Advances in Nutrition. 100710. 10.1016/j.advnut.2026.100710

2. **C. Swackhamer**✉, A. Hayes. "Something to chew on – oral processing simulators should be developed for integration with in vitro digestion workflows." 2026. Food Bioscience. 108271. 10.1016/j.fbio.2026.108271

3. **C. Swackhamer**†, S. Jang†, B.R. Park, B. R. Hamaker✉, S. K. Jung✉. "Structure-based standardization of prebiotic soluble dietary fibers based on monosaccharide composition, degree of polymerization, and linkage composition." 2025. Carbohydrate Polymers. 123949. 10.1016/j.carbpol.2025.123949

4. **C. Swackhamer**, R. Doan*, Y. Lu, N. Nitin, G. M. Bornhorst✉. "Fracture and rheological properties of standardized, model solid foods influence their breakdown mechanisms during in vitro gastric digestion with simulated peristalsis." 2025. Food Hydrocolloids. 111580. 10.1016/j.foodhyd.2025.111580

5. L. Trindade Paes, T. Cantu-Jungles, M. Aparecida de Campos Costa, **C. Swackhamer**, B. R. Hamaker, F. Augusto Ribeiro de Barros✉. "Influence of toasted sorghum flour phenolic compounds and dietary fibers on gut microbiota and short-chain fatty acid production." 2025. Food Chemistry. 10.1016/j.foodchem.2025.147606

6. V. Igwe, D. Smith✉, C. Mensah, **C. Swackhamer**. "Synergistic Enhancement of Corn Insoluble Dietary Fiber via Combined Radiofrequency Heating and Enzymatic Hydrolysis: Fermentability and Short Chain Fatty Acid (SCFA) Production." 2025. Journal of Food Science. 90:e70548. 10.1111/1750-3841.70548

7. P. Torres-Aguilar, A. Hayes, **C. Swackhamer**, L. Michelin, E. Ayua, V. Mugalavai, B. Hamaker✉. "Role of Habitual Diet in Metabolic Fuel Utilization and Metabolic Flexibility, Evidence in Kenyan and U.S. Cohorts." 2025. European Journal of Clinical Nutrition. 10.1038/s41430-025-01665-3

8. F. Cisse†, **C. Swackhamer**†, H. Diall, A. Rahmanifar, M. Sylla, A.R. Opekun, M.A. Grusak, A. H-M. Lin, E.A. Pletsch, A. M.R. Hayes, R. Quezado-Calvillo, B.L. Nichols, B. R. Hamaker✉, "Stunted African toddlers digest and obtain energy from energy-dense thick sorghum porridge." 2025. European Journal of Clinical Nutrition. 10.1038/s41430-025-01632-y

9. A. Hayes†✉, **C. Swackhamer**†, R. Quezado-Calvillo, N.F. Butte, E.E. Sterchi, B.L. Nichols✉, B.R. Hamaker✉. "Moderating carbohydrate digestion rate in mice promotes fat oxidation and metabolic flexibility revealed through a new approach to assess metabolic substrate utilization." 2025. European Journal of Nutrition. 10.1007/s00394-025-03585-1

10. **C. Swackhamer**, T. Bedane, S. Keppler, A. Poltorak, K. Cheung, N. Awais, F. Marra, G. Bornhorst✉, "Development and Analysis of a Multi-Module Peristaltic Simulator for Gastrointestinal Research." 2023. Food Research International. 112877. 10.1016/j.foodres.2023.112877.

11. **C. Swackhamer**, R. Doan*, G. Bornhorst✉, "Development and characterization of standardized model, solid foods with varying breakdown rates during gastric digestion." 2022. Journal of Food Engineering. 316:110827. 10.1016/j.jfoodeng.2021.110827.

12. Y. Mennah-Govela, **C. Swackhamer**, G. Bornhorst✉, "Gastric secretion rate and protein concentration impact intragastric pH and protein hydrolysis during dynamic in vitro gastric digestion." 2021. Food Hydrocolloids for Health. 10.1016/j.fhfh.2021.100027.

13. **C. Swackhamer**, J. Mullin✉, "Emergency Transition of Intro Communication and Design Course to Remote Teaching." 2021. American Society of Engineering Education Conference. 10.18260/1-2–37025. **Won best paper award in Design in Engineering Education Division:** Award announcement.

14. A. Hayes, F. Gozzi, A. Diatta, T. Gorissen, **C. Swackhamer**, S. Bellmann, B. Hamaker✉, "Some pearl millet-based foods promote satiety or reduce glycaemic response in a crossover trial." 2020. British Journal of Nutrition. 10.1017/S0007114520005036.

15. N. Nativ-Zeltzer, R. Ueha, Y. Nachalon, B. Ma, G. Pastenkos, **C. Swackhamer**, G. Bornhorst, M. Lefton-Greif, J. Anderson, P. Belafsky✉. "Inflammatory Effects of Thickened Water on the Lungs in a Murine Model of Recurrent Aspiration." 2020. The Laryngoscope. 10.1002/lary.28948.

16. A. Hayes, **C. Swackhamer**, Y. Mennah-Govela, M. Martínez, A. Diatta, G. Bornhorst✉, B. Hamaker✉. "Pearl millet (*Pennisetum glaucum*) couscous breaks down faster than wheat couscous in the Human Gastric Simulator, though has slower starch hydrolysis." 2020. Food & Function. 11: 111-122. 10.1039/C9FO01461F. **Selected as cover article, created original art for journal cover using Adobe Illustrator:** https://pubs.rsc.org/en/content/articlelanding/2020/fo/d0fo90003f.

17. **C. Swackhamer**, Z. Zhang, A. Taha, G. Bornhorst✉. "Fatty acid bioaccessibility and structural breakdown from in vitro digestion of almond particles." 2019. Food & Function. 10: 5174-5187. 10.1039/C9FO00789J

18. **C. Swackhamer**, G. Bornhorst✉. "Fracture properties of foods: Experimental considerations and applications to mastication." 2019. Journal of Food Engineering. 263: 213-226. 10.1016/j.jfoodeng.2019.07.002

19. D. Ciolkosz✉, R. Hilton, **C. Swackhamer**, H. Yi, V. Puri, D. Swomley, G. Roth, "Farm-scale biomass pelletizer performance for switchgrass pellet production." 2015. Applied Engineering in Agriculture. 31(4): 559-567. 10.13031/aea.31.10803

* undergraduate student mentored
† co-first authors
✉ corresponding author

Submitted manuscripts

1. L. Trindade Paes, Vinícius Parzanini Brilhante de São José, Júlia Mendes, **Clay Swackhamer**, Thaisa Cantu-Jungles, Leonara Martins Viana, Carolina Woelffel Silva, Graziela Domingues de Almeida Lima, Luciana Azevedo, Hércia Stampini Duarte Martino, Bruce R. Hamaker, Frederico Augusto Ribeiro de Barros✉. "Toasted Sorghum Flours Alter Oxidative Stress and Reduce Aberrant Crypt Foci in Early-Stage 1,2-Dimethylhydrazine-Induced Colon Carcinogenesis in Rats."

Manuscripts in preparation

1. **C. Swackhamer**, A. Hayes, T. Cantu-Jungles, B.R. Hamaker. "Shared dynamics of the gut microbiome drive consistent responses to fiber interventions." This work is related to items 3, 4, 5, and 11 in the presentations section.

2. **C. Swackhamer**, N. Bulut, A. Terekhov, T. Cantu-Jungles, B.R. Hamaker. "Fine structural differences in monosaccharide composition and glycosidic linkage profile of maize bran arabinoxylans influence utilization by gut microbes." This work is related to items 7, 8, 9, and 12 in the presentations section.

Research support

Federal competitive grants (received)

1. **USDA-NIFA-AFRI postdoctoral research fellowship** *August 2022 - August 2024*
   Role: Principal Investigator
   Project mentor: Dr. Bruce Hamaker (Purdue)
   Proposal title: "Effect of fiber structure on the taxonomic composition of the colonic microbiome."
   Amount: $190,000.
   Responsibilities: Initiated project ideas, developed research design and hypothesis, wrote proposal, conducted experiments, prepared annual reports, prepared manuscripts
   Award information

National-level competitive proposals (not funded)

1. **Pork Checkoff (National Pork Board)** *Submitted January 2026*
   Role: Co-investigator
   Other investigators: Dr. David Dallas (Principal Investigator)
   Proposal title: "Exploring pork as an ideal source of digestible protein and bioactive peptides for the elderly."
   Responsibilities: Initiated project ideas, developed proposal (4 page letter of intent)

2. **Dairy Management INC (DMI product research call)** *Submitted October 2025*
   Role: Principal Investigator
   Other Investigators: Dr. David Dallas and Dr. Luyao Ma (Oregon State), Dr. Thomas Schumacher (Portland State University)
   Proposal title: "Decision support for optimal cheddar cheese ripening using ultrasonic testing and AI modeling."
   Responsibilities: Initiated project ideas, recruited team of researchers with the skillsets needed to carry out the project, collected preliminary data in the laboratory (ultrasound testing of cheese), created graphical abstract/summary figure for the proposal, wrote and submitted a full invited proposal based on competitive pre-proposal.

Federal competitive proposals (not funded)

1. **NIH-NICHD (R01)** *Submitted August 2025*
   Role: Contributor
   Principal Investigators: Dr. David Dallas (Oregon State) and Dr. Brian Scottoline (Oregon Health and Science University)
   Proposal title: "Pancreatic enzyme supplementation to enhance nutrient digestion, absorption and growth in preterm infants."
   Responsibilities: Collected preliminary data in the laboratory (in vitro digestions), prepared one of the three specific aims, conducted statistical analysis and visualization of preliminary data, created graphical abstract/summary figure for the proposal

2. **NSF Postdoctoral Research Fellowship (NSF 20-602)** *Submitted November 2020*
   Role: Principal Investigator
   Project mentor: Dr. Bruce Hamaker (Purdue)
   Proposal title: "Effect of fiber structure on the taxonomic composition of the colonic microbiome: Investigating the rules of life that govern substrate utilization by human-symbiotic microbes."
   Responsibilities: Initiated project ideas, developed research design and hypothesis, wrote proposal, prepared supporting documents

3. **USDA-NIFA-AFRI Predoctoral Fellowship (A7101)** *Submitted July 2019*
   Role: Project Director
   Project mentor: Dr. Gail Bornhorst
   Proposal title: "Development of model solid foods to elucidate the relationship between food breakdown rate and bioaccessibility of nutrients in the gastric environment."

Responsibilities: Developed research design and hypothesis, wrote proposal, prepared supporting documents

4. **NSF Graduate Research Fellowship Program (GRFP)** *Submitted October 2016*
   Role: Project Director
   Project mentor: Dr. Gail Bornhorst
   Proposal title: "Characterization and validation of particle collision rates during peristaltic flow using a computational fluid dynamic approach."
   Responsibilities: Developed research design and hypothesis, wrote proposal, prepared supporting documents

State-level proposals (received/recommended for funding)

1. **Oregon Wheat Board** *Submitted January 2026*
   Role: Co-principal investigator
   Other co-principal investigators: Dr. Anna Hayes, Dr. David Dallas
   Proposal title: "Elucidating human gut health benefits of Oregon-grown whole grain wheat."
   Amount: $27,700
   Responsibilities: Helped develop project ideas, contributed to proposal (5 page full proposal), co-presented the proposal at an in-person meeting of the Oregon Wheat Board with Dr. Anna Hayes.

State-level proposals (not funded)

1. **Oregon Hazelnut Commission** *Submitted February 2026*
   Role: Co-investigator
   Other investigators: Dr. David Dallas (Principal Investigator)
   Proposal title: "Identifying gut health benefits of hazelnuts."
   Responsibilities: Initiated project ideas, developed proposal (4 page full proposal), co-presented the proposal at a meeting of the Oregon Hazelnut Commission with Dr. David Dallas.

2. **Oregon Potato Commission** *Submitted February 2026*
   Role: Co-investigator
   Other investigators: Dr. David Dallas (Principal Investigator)
   Proposal title: "Establishing the gut health benefits of Oregon potatoes through analysis of digestion-derived bioactive peptides and effects on the microbiome."
   Responsibilities: Initiated project ideas, developed proposal (5 page full proposal)

Institutional-level funding received

1. **UC Davis Jastro-Shields Fellowship** for graduate students in agricultural or environmental sciences based on competitive application for original research. Proposal title: "Influence of mechanical properties on the particle breakdown dynamics of standard, model solid foods during in vitro gastric digestion using a peristaltic simulator." $1,000. *July 2021.*

2. **UC Davis Jastro-Shields Fellowship** for graduate students in agricultural or environmental sciences based on competitive application for original research. Proposal title: "Breakdown of model solid foods based on the Standard American Diet using shear and fracture testing diet to quantify mechanical breakdown during in vitro gastric digestion." $3,000. *September 2020.*

3. **UC Davis Graduate Student Travel Award** $500. *May 2020.*

4. **UC Davis Biological Systems Engineering Graduate Program Fellowship Award** $2,000. *May 2020.*

5. **UC Davis Walter Rosenburg Research Fund** to support the research of PhD candidates in Biological and Agricultural engineering, based on faculty nomination $2,000. *May 2020.*

6. **UC Davis John C. Harper Memorial Fellowship** for graduate students in Biological Systems Engineering specializing in Food Engineering, based on faculty nomination $3,500. *November 2019.*

7. **UC Davis Jastro-Shields Fellowship** for graduate students in agricultural or environmental sciences based on competitive application for original research. Proposal title: "Development and mechanical property testing of a novel system of test foods for studying food breakdown during in vitro gastric digestion." $2,500. *July 2019.*
8. **UC Davis Jastro-Shields Fellowship** for graduate students in agricultural or environmental sciences based on competitive application for original research. Proposal title: "Development and validation of a low-cost, single camera calibrated computer vision system for particle size analysis of foods subjected to in vitro digestion." $2,000. *October 2018.*
9. **UC Davis Jastro-Shields Fellowship** for graduate students in agricultural or environmental sciences based on competitive application for original research. Proposal title: "Characterizing the influence of physical properties on lipid release from fractured almonds using simulated gastric digestion." $1,700. *October 2017.*
10. **UC Davis Mentoring at Critical Transitions Award** for graduate students based on competitive application and potential to transition to future role as mentor, $3,000. *September 2017.*
11. **UC Davis Scholars Award** first year fellowship for outstanding graduate students at UC Davis, $58,000. *September 2016.*

Industry Experience

Engineering intern *Winter-Summer 2016*
LignoLink, INC.
Innovation Park, PA

- Conducted aqueous ammonia pretreatment and enzymatic hydrolysis reactions on 98 biomass samples (corn stover)
- Designed, constructed, and validated High Performance Liquid Chromatography (HPLC) assay for 5 sugar monomers in hydrolyzed biomass samples
- HPLC development included equipment selection, purchase, installation, calibration, and development of semi-automated data analysis routine using MATLAB
- Created the first logo and website for a faculty-led startup company, funded by the National Science Foundation-Small Business Innovation Research (NSF-SBIR) program
- Information about the company: NSF page, News article

Engineering intern *Summer 2015*
McCormick & Company INC, Materials Process Engineering Group
Hunt Valley, MD

- Helped create plan to achieve $160,000 annual cost savings by reducing overblending, which can affect product bulk density and lead to overfill losses
- Measured physical properties of dry seasoning mixes (gravy mixes, taco seasoning, steak seasonings) using 5 instruments in food powder technology platform
- Conducted over 300 tests in the laboratory, mostly involving powder flowability and compaction properties; processed data using descriptive statistics and modeling
- Quantified link between blend time and product bulk density
- Collaborated with McCormick engineers in 3 countries

Teaching Experience

Co-instructor of record *Fall 2025*
Course: FST 422/522: Food Chemistry Fundamentals
Oregon State University Department of Food Science

- Delivered the carbohydrate chemistry section of the course (to 18 students)

- Responsibilities included giving 6 lectures (50 minutes each), developing and grading one homework assignment, and holding a review session for the carbohydrate related course material

Short course instructor *October 2023*
Module title: "Computational Approaches to Big Data in Gut Microbiome Studies"
Whistler Center for Carbohydrate Research Annual Short Course
Purdue University Department of Food Science

- Co-designed a short course module with Dane Deemer, Dr. Bruce Hamaker, and Dr. Stephen Lindemann
- Developed instructional materials including a (hopefully fun) demonstration using a deep convolutional neural network for image classification of hand-drawn artwork submitted by Whistler Center members
- Taught the first half of the module (45 min)
- Helped answer questions from industry members (about 25 food industry professionals attended the module)

Teaching assistant *Spring 2019, Fall 2019, Winter 2020, Spring 2020, Fall 2020, Winter 2021, Spring 2021, Fall 2021, Winter 2022*
Course ENG 003: Intro to Engineering Design
UC Davis College of Engineering

- Responsible for two lab sections every academic quarter (ca. 22 students each) for 9 academic terms, working with about 350 students in total
- Taught basics of circuit building using Arduino and Raspberry Pi
- Topics included circuit prototyping on breadboards, writing Arduino and Python code, interfacing with analog devices, and troubleshooting circuits using a multimeter
- Built circuits to interface with sensors measuring temperature, weight, light, moisture, color, humidity, dust level, carbon dioxide concentration, pH, and water flow rate
- Built circuits to interface with devices such as servo motors, DC motors, solenoid valves, and cameras
- Taught engineering design topics including the iterative process of design, prototyping, decision matrices, and effective presentation techniques
- Directly advised ca. 90 student teams as they created original design projects, including prototypes for detecting pests in greenhouses (by counting insects stuck to sticky tape with a digital camera and image processing algorithm), water quality testing (using conductivity), and grape harvesting decision support (by testing °brix)
- Participated in the emergency transition of this highly in-person course to remote teaching in the spring of 2020 during the COVID-19 pandemic. Conducted research on student learning outcomes using a validated survey instrument. Published peer-reviewed paper on our findings at the American Society of Engineering Education conference and won the best paper award in the division. Award announcement.

Teaching assistant *Fall 2018*
Course EBS 001: Foundations of Biological Systems Engineering
UC Davis Department of Biological and Agricultural Engineering

- Assisted with construction of seven algae photobioreactors and seven solar food dehydrators prototypes made by student teams for their final project
- Provided written feedback for 49 students on homeworks and exams
- Held weekly office hours

Teaching intern *Fall 2015*
Course BE 301: Mathematical Modeling of Biological and Physical Systems
Penn State Department of Agricultural and Biological Engineering

- Teaching intern for course BE 301: Mathematical Modeling of Biological and Physical Systems (approximately 20 students)
- Ran review sessions and helped troubleshoot student MATLAB codes
- Held weekly office hours

Mentorship

- **Research mentor** for Rebecca Houlehan, undergraduate researcher in the Dallas laboratory at Oregon State University (2025-2026). Her project is ongoing.
- **Thesis committee member** for Daisy Johnson, undergraduate honors scholar at Oregon State University majoring in Biohealth Sciences (HBS). Thesis title: "Development of SHIME protocol in Preterm Infants." Defense date: 06/03/2025
- **Thesis committee member** for Hailey Renee Kelly, undergraduate honors scholar at Oregon State University majoring in Biohealth Sciences (HBS). Thesis title: "Characterizing the Immunomodulatory Effects of Peptides from Processed Donor Human Milk in Macrophage Cell Models: Implications for Preterm Infant Immune Response." Defense date: 12/04/2025
- **Research mentor** for Raymond Doan, undergraduate researcher in the Bornhorst laboratory at UC Davis. Co-authored a poster at the undergraduate research exposition (04/19/2021): "Buffering Capacity of Standard Model Foods as Related to Simulated Gastric Digestion."
  Additionally, co-authored two peer-reviewed journal articles together:
  – Swackhamer et al., (2025): https://doi.org/10.1016/j.foodhyd.2025.111580
  – Swackhamer et al., (2022): https://doi.org/10.1016/j.jfoodeng.2021.110827
- **Research mentor** for Zhe (Kelly) Fu, undergraduate researcher in the Bornhorst laboratory at UC Davis. Co-authored a poster at the undergraduate research exposition (04/24/2019): "Effect of simulated gastric juice on texture change of almond particles during simulated gastric digestion."
- **Research mentor** for Xiaohan (Hazel) Cao, undergraduate researcher in the Bornhorst laboratory at UC Davis. Co-authored a poster at the undergraduate research exposition (04/15/2018): "Experimental investigation of the impact of processing variables on fracture properties of almonds."

Presentations

1. E. Moncada, W-J Choung, **C. Swackhamer**, B. L. Nichols, A. M. R. Hayes. "Blunting starch digestion impairs metabolic flexibility in mice." Poster presentation. 33rd Annual Meeting of the Society for the Study of Ingestive Behavior. August 2026.

2. I. M. Lopez-Rodulfo, **C. Swackhamer**, J. Bechtner, F. Fang, B. R. Hamaker, M. M. Martinez. "Polyphenols modulate faecal fermentation of dietary fibre depending on polysaccharide chemistry and supramolecular assembly." Poster presentation. 18th International Hydrocolloids Conference. March 2026.

3. **C. Swackhamer**, A. M. R. Hayes, T. Cantu-Jungles, B. R. Hamaker. "Evidence for consistent response to dietary fibers using mathematical modeling of the dynamics of the colonic microbiome." Oral presentation. Starch Digestion Consortium Meeting. November 2025.

4. **C. Swackhamer**, A. M. R. Hayes, T. Cantu-Jungles, B. R. Hamaker. "Meta analysis reveals that prebiotic dietary fiber interventions may lead to more consistent responses in the microbiome than non-fermentable placebo." Poster presentation. Linus Pauling Institute Diet and Optimum Health Conference. September 2025.

5. **C. Swackhamer**, T. Cantu-Jungles, B. R. Hamaker. "Shared dynamics of the colonic microbiome make responses to dietary fiber interventions unexpectedly consistent." Poster presentation. American Society of Nutrition Conference. June 2025. Abstract in Current Developments in Nutrition

6. A. Hayes[†], **C. Swackhamer**[†], R. Quezado-Calvillo, N. F. Butte, E. E. Sterchi, B. L. Nichols, B. R. Hamaker "Diets with higher slowly digestible starch improve metabolic flexibility using a new assessment approach." 2025. Poster presentation. American Society of Nutrition Conference. June 2025. Abstract in Current Developments in Nutrition

7. Rajashri Lun, T. Cantu-Jungles, **C. Swackhamer**, B. R. Hamaker. "Synergistic effect of dietary fibers and nano-encapsulation of spice oleoresins: a novel gut-brain axis approach for depression

management." Poster presentation. Whistler Center for Carbohydrate Research Annual Meeting Technical Workshop. *April 2025.*

8. **C. Swackhamer**, T. Cantu-Jungles, N. Bulut, A. Terekhov, B. R. Hamaker. "Engineering the gut microbiome using designer prebiotics and consumer-resource modeling" Oral presentation. Indiana Nutrition for Health Research Showcase. *February 2025.*

9. **C. Swackhamer**, N. Bulut, T. Cantu-Jungles, B. R. Hamaker. "Engineering the gut microbiome using designer prebiotics and consumer-resource modeling." Oral presentation. Conference of Food Engineering (CoFE). *August 2024.*

10. **C. Swackhamer**. "Effect of fiber structure on the taxonomic composition of the colonic microbiome." Oral presentation. USDA project directors' meeting. *May 2024.*

11. **C. Swackhamer**, T. Cantu-Jungles, B. R. Hamaker. "Three patterns emerge in the gut microbial community during in vitro fecal fermentation of dietary fibers varying in their level of complexity." Poster presentation. Midwest Microbiome Symposium. *May 2024.*

12. **C. Swackhamer**. "Engineering all along the digestive tract: model foods with designer breakdown kinetics, multiphysics CFD modeling, and microbiome engineering." Oral presentation. Michigan State University Department of Biosystems and Agricultural Engineering Seminar. *March 2024.*

13. **C. Swackhamer**. "Effect of Arabinoxylan Structure on the Microbiota during in vitro Fecal Fermentation." Oral presentation. Whistler Center for Carbohydrate Research Update Seminar Series. Purdue Department of Food Science. *September 2023.*

14. **C. Swackhamer**. "Artificial Intelligence in Microbiome Research: Concepts and Applications to the Food Industry." Oral presentation. Whistler Center for Carbohydrate Research Annual Meeting Technical Workshop. Purdue Department of Food Science. *May 2023.*

15. **C. Swackhamer**, B. Hamaker. "Project update: Effect of fiber structure on the taxonomic composition of the colonic microbiome." Poster presentation. Whistler Center for Carbohydrate Research Annual Meeting. Purdue Department of Food Science. *May 2023.*

16. **C. Swackhamer**, R. Doan*, G. Bornhorst. "Breakdown Mechanisms of Model Solid Foods During in Vitro Gastric Digestion with Simulated Peristalsis." Oral presentation. Conference of Food Engineering. *September 2022.*

17. **C. Swackhamer**, R. Doan*, G. Bornhorst. "Food physical properties influence mechanisms of particle breakdown of solid foods during in vitro gastric digestion in the Human Gastric Simulator." Poster presentation. Institute of Food Technologists Conference. *July 2022.*

18. **C. Swackhamer**, "Breakdown of solid foods during in vitro gastric digestion." PhD exit seminar in UC Davis Department of Biological and Agricultural Engineering. *January 2022.*

19. **C. Swackhamer**, J. Mullin "Emergency Transition of Intro Communication and Design Course to Remote Teaching." Remote/virtual oral presentation. American Society of Engineering Education Conference. **Won best paper award in the division.** Award announcement. *July 2021.*

20. **C. Swackhamer**, R. Doan*, G. Bornhorst. "Fracture property analysis of standardized, model solid foods during in vitro gastric digestion." Remote/virtual poster presentation. Institute of Food Technologists Conference. *July 2021.*

21. R. Doan*, **C. Swackhamer**, G. Bornhorst. "Buffering capacity of standard model foods as related to simulated gastric digestion." Virtual poster presentation. UC Davis Undergraduate Research Conference. Davis, CA, *April 2021.*

22. **C. Swackhamer**, S. Keppler, A. Poltorak, T.F. Bedane, K. Cheung, N. Awais, F. Marra, G. Bornhorst. "Video analysis and computational fluid dynamic modeling of a novel multi-module peristaltic simulator for gastrointestinal research." Remote/virtual poster presentation. Institute of Food Technologists Conference. **Won first place in Food Engineering Division graduate student research poster competition.** *July 2020.* Award announcement.

23. **C. Swackhamer**, R. Doan*, G. Bornhorst. "Development and characterization of standardized model, solid foods with varying breakdown rates during gastric digestion." Remote/virtual poster presentation. Institute of Food Technologists Conference. *July 2020.*

24. Z. Fu*, **C. Swackhamer**, G. Bornhorst. "Effect of simulated gastric juice on texture change of almond particles during simulated gastric digestion." Poster presentation. UC Davis Undergraduate Research Conference. Davis, CA, *April 2019.*

25. **C. Swackhamer**, Z. Zhang, A. Taha, G. Bornhorst. "Fatty acid bioaccessibility, structural breakdown, and properties of digesta from in vitro digestion of almond particles." Poster presentation. Conference of Food Engineering. Minneapolis, MN, *September 2018.*

26. A. Hayes, **C. Swackhamer**, M. Martínez, Y. Mennah-Govela, G. Bornhorst, B. Hamaker. "Breakdown rate of couscous made from pearl millet versus wheat in a simulated gastric environment linked to gastric emptying." Poster presentation. Institute of Food Technologists Conference. Chicago, IL, *July 2018.*

27. X. Cao*, **C. Swackhamer**, G. Bornhorst. "Experimental investigation of the impact of processing variables on fracture properties of almonds." Poster presentation. UC Davis Undergraduate Research Conference. Davis, CA, *April 2018.*

28. A. Hayes, **C. Swackhamer**, M. Martínez, Y. Mennah-Govela, G. Bornhorst, B. Hamaker. "Insights to the delayed gastric emptying rate and slow digestibility of pearl millet couscous." Poster presentation. Sorghum in the 21st Century. Cape Town, South Africa, *April 2018.*

29. A. Olenskyj, Y. Mennah-Govela, **C. Swackhamer**, K. Rios-Villa, G. Bornhorst. "Softening half-time and final normalized hardness as indicators of food structural breakdown during in vitro digestion." Poster presentation. Institute of Food Technologists Conference. Las Vegas, NV, *June 2017.*

30. **C. Swackhamer**, Y. Mennah-Govela, G. Bornhorst. "Physical property changes in soaked and activated almonds." Poster presentation. Almond Board of California Conference. Sacramento, CA, *December 2016.*

31. **C. Swackhamer**. "Effect of codon optimization on bacterial translation elongation rates." Presentation. Penn State Department of Agricultural and Biological Engineering Department seminar: Exit seminar for undergraduate honors thesis presentation. University Park, PA, *December 2015.*

32. **C. Swackhamer**, "Industrial batch fermentation of L-Lysine using *Corynebacterium glutamicum*." Presentation. Northeast Agricultural and Biological Engineering Conference (ASABE regional conference), Newark, DE, *July 2015.*

33. **C. Swackhamer**, A. Smith, E. Sileo, S. Krug, "Next-generation approaches to overcome the challenges of metabolic pathway engineering." Presentation. iGEM Giant Jamboree, Boston, MA, *October 2014.*

34. **C. Swackhamer**, B. Wright, "Small scale densification of *Panicum virgatum* through pelletization." Poster presentation. Northeast Agricultural and Biological Engineering Conference (NABEC), Altoona, PA, *July 2013.*

35. **C. Swackhamer**, B. Wright, "Small scale densification of *Panicum virgatum* through pelletization." Poster presentation. Penn State Undergraduate Research Exposition, University Park, PA, *April 2013.*

36. **C. Swackhamer**, V. Vadyak, N. Vitacco, A. Alsuwaidi, "Portable recharging system for personal electronic devices." Poster presentation. Penn State Engineering Design Showcase, University Park, PA, *December 2012.*

* undergraduate student mentored
† co-first authors

Awards

1. **Best paper award** in Design in Engineering Education Division at American Society of Engineering Education Conference. *July 2021.* Award announcement.
2. **Best teaching assistant award 2020**. UC Davis College of Engineering. *August 2020.* Award announcement.
3. **First place in graduate student research poster competition**. Food Engineering Division. Institute of Food Technologists Conference. *July 2020.* Award announcement.
4. **Outstanding Senior Award** in Penn State Department of Biological and Agricultural Engineering. *May 2016.*
5. **First place** in undergraduate paper competition for presentation of "Industrial Batch Fermentation of L-Lysine using *Corynebacterium glutamicum*." Northeast Agricultural and Biological Engineering Conference (NABEC). *July 2015.*
6. **Gold medal** for team presentation of "Next-Generation Approaches to Overcome the Challenges of Metabolic Pathway Engineering." iGEM Giant Jamboree. *October 2014.*
7. **Myriant Corporation Scholarship** for Excellence in Bioenergy. *October 2013.*
8. **Third place award** in engineering division for poster presentation of "Small Scale Densification of *Panicum virgatum* through Pelletization." Penn State Undergraduate Research Exposition. *April 2013.*
9. **"Best in class" award** for poster presentation of "Portable Recharging System for Personal Electronic Devices." Penn State Engineering Design Showcase. *December 2012.*

Professional trainings and certifications

1. **Biosafety for Principal Investigators, Lab Personnel, and IBC Members**. Two hour course with quiz and certificate, developed by the Collaborative Institutional Training Initiative (CITI). *2023.*
2. **Responsible Conduct of Research Training for Faculty, Postdoctoral, and Graduate Students**. Two hour course with quiz and certificate, developed by the Collaborative Institutional Training Initiative (CITI). *2023.*
3. **Social and Behavioral Research Best Practices for Clinical Research**. Three hour course with quiz and certificate, developed by the Collaborative Institutional Training Initiative (CITI). *2020.*
4. **Social and Behavioral Science Responsible Conduct of Research**. Three hour course with quiz and certificate, developed by the Collaborative Institutional Training Initiative (CITI). *2020.*
5. **Proposal Writing Program: Writing Successful Grant Proposals.** Five hour professional development program with quiz and certificate, developed by the American Association for the Advancement of Science (AAAS). *2017.*

6. **Synthetic biology certificate for undergraduates**, Synthetic Biology Engineering Research Center (SYNBERC). *2015.*
7. **Laboratory safety fundamentals** certificate, UC Davis Safety Services. *2016-2022.*
8. **Passed Fundamentals of Engineering Exam** (FE-general disciplines). Certified engineer in Training (EIT), license number ET021732. *2015-2021.*

Professional Affiliations and Leadership

1. **Member**, Phi Tau Sigma, The Honor Society of Food Science and Technology *(2025-present)*
2. **Past president and co-founder**, Society of Food Engineering Student Division *(2020-2021)*
3. **Member**, Sigma Xi, The Scientific Research Honor Society *(2020-present)*
4. **Representative for UC Davis General Graduate Student Association**, Biological Engineering Graduate Student Association, UC Davis *(2019-2020)*
5. **Secretary**, Biological Engineering Graduate Student Association, UC Davis *(2018)*
6. **Member**, Society of Food Engineering (SoFE) *(2018-present)*
7. **Alumni Relations Co-Chair**, Biological Engineering Graduate Student Association, UC Davis *(2017)*
8. **Laboratory Safety Officer**, Food Engineering Laboratory, UC Davis *(Spring 2017, Spring 2019)*
9. **Member**, Institute of Food Technologists (IFT) *(2016-present)*
10. **Treasurer**, Alpha Epsilon, Honors Society of Biological Engineering, Penn State chapter *(2015)*
11. **Member**, American Society of Agricultural and Biological Engineers (ASABE) *(2015-present)*
12. **Alumni Relations Chair**, Alpha Gamma Rho National Agricultural Sciences Fraternity *(2013-2014)*
13. **Financial Chair**, Penn State Agricultural Sciences Student Council *(2013-2014)*
14. **Member**, Gamma Sigma Delta, National Honors Society in Agriculture *(2013-present)*

Service

Member of ASABE committee on food processing *2025 - present*

- Member of technical committee on food processing standards (PRS-703)

Service as NCEES committee member *Spring 2023*

- Invited to serve on the Agricultural and Biological Engineering PE Exam Standard Setting Committee
- Evaluated potential questions for the PE exam
- Completed basic training in exam psychometrics, helped determine the cut score as part of the committee

Service as peer reviewer *2022-present*
30 manuscripts reviewed across 16 scientific journals

Journal name	Number of manuscripts reviewed
Journal of Food Science	9
Computers and Electronics in Agriculture	3
Applied Sciences	3
Gels	2
Food Research International	2
npj Science of Food	1
Nutrition Research	1
Journal of Texture Studies	1
Food Structure	1
Frontiers in Nutrition	1
Processes	1
Fluids	1
Information	1
iMeta	1
BMC Genomics	1
JDS Communications	1

Miscellaneous projects and outputs

1. J. Coupland, M. James, M. Jiaqi, P. Kumar Sappati, **C. Swackhamer**, O. Tasie. "The First Graduating Class of IFT's Editorial Scholars Program." 2026. Co-authored an editorial in the Journal of Food Science about the learnings from being part of the first graduating class of the Institute of Food Technologists (IFT) editorial scholars program. PDF available online

2. **C. Swackhamer** "Re-analysis of Environmental fluctuations reshape an unexpected diversity-disturbance relationship in a microbial community." 2023. Project report for course ANSC 516 (Molecular microbiome analysis) I audited in the Department of Animal Science at Purdue University in Spring 2023. PDF on Github

3. **C. Swackhamer**, "Solving the heat equation for a slice of pie using Green's functions." 2017. Project report for course ECH 259 (Advanced engineering mathematics) in the Department of Chemical Engineering at UC Davis. PDF on Github.

4. **C. Swackhamer**, "Effect of codon optimization on bacterial translation elongation rates." 2015. Undergraduate honors thesis in the Department of Biological and Agricultural Engineering at Penn State. PDF on Github.

5. Codes to put a grid on an image or trace an object in an image using MATLAB: file exchange. 2018.

6. YouTube video: "AI-assisted microbiome data analysis using chat GPT". 2025